

# THE ZERO PARADOX

*ZP-C: Information Theory*

*Initial: April 2026 | Current: 1.25, October 2026*

This document is self-contained within information theory and discrete analysis on  $Q_2$ , with two named cross-layer dependencies. The topological structure of  $Q_2$  — total disconnectedness (ZP-B T5), the clopen ball hierarchy, and the binary existence axiom (AX-B1) — is imported from ZP-B. The Candidate Theorem T-BUF (Section V) additionally invokes ZP-A's state-transition definition D2 ( $f(x) = x \vee \alpha$ ) to identify the resulting state as the Binary Snap. T-BUF is conditional on execution: its Step 2, that the configuration at  $P_0$  is executing, is not derived here or in ZP-E; it is the occurrence commitment (ZP-E § I-DA1), which DA-1 consumes and does not supply. Every claim is marked as Derived, Axiomatic, Defined, or Candidate.

Encoding commitment. The two ontological states of AX-B1 are represented here as point-mass (Dirac) distributions over  $\{0, 1\}$  — the minimum-sufficient probabilistic encoding (RP-1, Section II). Under this encoding, the information-theoretic separation between  $\perp$  and the minimum nonzero state ( $\epsilon_0$ ) is exactly 1 bit (T1b). The 1-bit result is conditional on this encoding; RP-1 declares and justifies the commitment. Separately, at  $\perp$ , the primary descriptive tool of information theory — surprisal — is unbounded above (L-INF, Section III): no finite external description can contain  $\perp$ . These two results — the 1-bit cost of the transition and the infinite descriptive cost of the origin — are the information-theoretic expression of the same structural constraint on  $\perp$ .

Illustrated Companion: A paired ZP-C Illustrated Companion provides concrete examples and visual intuitions for the results here. Examples are kept separate from the formal layers to distinguish illustrative material from proofs.

## I. Kolmogorov Complexity and the Incompressibility Threshold

Let  $x$  be a binary string of length  $n$ . The conditional Kolmogorov complexity is:

$$K(x|n) = \min \{ |p| : U(p, n) = x \}$$

where  $U$  is a fixed universal Turing machine and  $|p|$  is the length of program  $p$ .

### Definition D1 — Incompressibility Threshold

$$P_0 = \lim_{n \rightarrow \infty} K(x|n) / n$$

$P_0$  is the algorithmic entropy rate of  $x$ . When  $P_0$  reaches its maximum,  $x$  has exhausted its capacity for self-description.

### Remark R1 — Scope of $P_0$

What  $P_0$  establishes (Derived): the point at which  $x$  becomes incompressible.

What  $P_0$  does not establish: that incompressibility causes the Binary Snap. The generative claim is a Candidate Theorem (T-BUF), not a bare axiom, and it is conditional on execution at  $P_0$ . See Section V.

## II. Informational Work: State Representations and JSD

### 2.1 The Representation Principle

#### Principle RP-1 — Minimum Sufficient Probabilistic Representation

The minimum sufficient probabilistic representation of a binary ontological state is a point-mass (Dirac) distribution over  $\{0,1\}$ : all probability mass is assigned to the value the state occupies, and none to the other value.

Justification: A point-mass distribution is the unique distribution that is (a) faithful — assigns non-zero probability only to the ontologically actual value, and (b) non-redundant — carries no uncertainty about which state is occupied. Any distribution with mass at both values represents partial existence and partial non-existence simultaneously. AX-B1 excludes this.

Status: PRINCIPLE — representational commitment parallel to MP-1 in ZP-B.

#### Theorem T1 — State Representations are Uniquely Derived

Given AX-B1 and RP-1, the distributions are unique:

$P = (1, 0)$  (Null State: all mass at 0 — non-existence)

$Q = (0, 1)$  (Minimum Nonzero State: all mass at 1 — existence)

Proof: AX-B1 establishes two ontological states. RP-1 requires point-mass representation of each. The Null State occupies value 0:  $P = (1,0)$ . The minimum nonzero state occupies value 1:  $Q = (0,1)$ . No distribution  $(p, 1-p)$  with  $0 < p < 1$  is consistent with AX-B1.  $P$  and  $Q$  are unique. Status: DERIVED from AX-B1 and RP-1. ✓

#### Corollary T1b — JSD = 1 bit

For  $P=(1,0)$  and  $Q=(0,1)$  with  $M=(1/2, 1/2)$ :

$D_{KL}(P||M) = 1$  bit,  $D_{KL}(Q||M) = 1$  bit,  $JSD(P||Q) = 1$  bit

$E = JSD(P||Q) = 1$  bit [information-theoretic, dimensionless]

Status: Derived from AX-B1 and RP-1 via T1.  $E$  is not physical energy in joules. Dimensional bridge belongs to ZP-E as BA-1.

#### Definition D3 — Dirac Measure $\delta_0$

$$\int_{\Omega} f d\delta_0 = f(0)$$

$\delta_0$  places unit mass at 0. Compatible with the discrete topology of  $\{0,1\}$  and with the totally disconnected topology of  $Q_2$  (ZP-B T5).  $\delta_0$  governs behaviour exactly at  $x = 0$ .

#### Remark R2 — Hamming Cross-Validation

$$d_H(P, Q) = 1 \text{ [Hamming, on } \{0,1\}]$$

#### Remark R2 — Hamming Cross-Validation

The agreement between  $d_H = 1$  and  $E = 1$  bit is a consistency check, not a proof that Hamming distance and JSD are equivalent in general. Both are computed by independent methods on the same AX-B1-derived distributions.

### III. The Discrete Surprisal Field on $Q_2$

#### Remark R3 — Why the Smooth Embedding Remains Retired

ZP-B T5 establishes  $Q_2$  is totally disconnected. ZP-B T2 establishes every ball is clopen. Importing smooth calculus operators onto a totally disconnected space imports a smoothness assumption that contradicts ZP-B. The smooth embedding, MO-1, and P1 from v1.1 remain retired.

#### Definition D4 — Discrete Surprisal Function I

For  $x \in Q_2$  with  $x \neq 0$  and  $P(x) > 0$ :

$$I(x) = -\log_2 P(x)$$

As  $v_2(x) \rightarrow +\infty$  ( $x$  approaches 0 in the 2-adic metric),  $I(x) \rightarrow +\infty$ : states 2-adically close to 0 are informationally extreme.

The probability measure  $P$  on  $Q_2 \setminus \{0\}$  is the branching measure induced by the binary ball hierarchy of  $Q_2$  (RP-2 below): at each level  $k$ , the two sub-balls of  $B(0, 2^{-k})$  each receive half the probability mass of their parent ball.

#### Principle RP-2 — Branching Measure on $Q_2 \setminus \{0\}$

The probability measure  $P$  on  $Q_2 \setminus \{0\}$  is the canonical branching measure induced by the binary ball hierarchy: at each level  $k$ , each sub-ball receives half the probability mass of its parent. This is a representational commitment — a specific measure choice not uniquely forced by AX-B1 alone.

Justification: The branching measure assigns equal weight to each binary branch, consistent with AX-B1's symmetric treatment of existence and non-existence. Alternative measures (e.g. non-uniform branching weights) would depart from this symmetry without additional motivation.

Parallel: RP-1 fixes the point-mass representation of ontological states. RP-2 fixes the probability measure on the 2-adic domain. Both are representational commitments sitting alongside, not derived from, AX-B1.

Status: PRINCIPLE — representational commitment. Required by T2 and L-INF.

#### Definition D5 — Discrete Surprisal Difference Operator DF

For  $x, y \in Q_2 \setminus \{0\}$ :

$$DF(x, y) = I(y) - I(x) = \log_2 [P(x) / P(y)]$$

DF is antisymmetric:  $DF(x, y) = -DF(y, x)$ . No smoothness is assumed; DF is defined entirely pointwise.

### Definition D6 — Discrete Circulation (Extended)

FINITE CASE: A discrete loop  $\gamma_n = (x_0, x_1, \dots, x_n, x_0)$  in  $Q_2 \setminus \{0\}$  returning to its start. Circulation:  $C(\text{DF}, \gamma_n) = \sum_{i=0}^n \text{DF}(x_i, x_{i+1})$  where  $x_{n+1} = x_0$ .

Note: By telescoping,  $C(\text{DF}, \gamma_n) = I(x_0) - I(x_0) = 0$  for all finite loops. DF is conservative on all finite loops.

INFINITE CASE: An infinite sequence  $\sigma = (x_1, x_2, \dots)$  in  $Q_2 \setminus \{0\}$  where  $v_2(x_i) = i$  for all  $i \geq 1$ . Partial circulation:  $C_n(\text{DF}, \sigma) = I(x_{n+1}) - I(x_1)$ .

Circulation of  $\sigma$ :  $C(\text{DF}, \sigma) = \lim_{n \rightarrow \infty} C_n(\text{DF}, \sigma)$  if the limit exists or diverges.

An infinite sequence  $\sigma$  exhibits non-conservative behaviour if  $C(\text{DF}, \sigma)$  diverges.

### Theorem T2 — DF Exhibits Non-Conservative Behaviour on Infinite Sequences Approaching 0

DF is conservative on all finite loops ( $C = 0$  by telescoping). On infinite sequences through the ball hierarchy approaching 0, the circulation diverges:  $C(\text{DF}, \sigma) \rightarrow +\infty$ .

Proof: Let  $\sigma = (x_1, x_2, \dots)$  with  $x_i = 2^{-i}$ . Then  $P(x_i) = 2^{-i}$ , so  $I(x_i) = i$ . Partial circulation  $C_n = I(x_{n+1}) - I(x_1) = n$ . As  $n \rightarrow \infty$ :  $C(\text{DF}, \sigma) = +\infty$ .

Divergence arises from the unbounded depth of the 2-adic ball hierarchy (ZP-B). Finite conservation and infinite divergence are distinct regimes; no contradiction arises.

Status: DERIVED from ZP-B ball hierarchy structure and branching measure on  $Q_2$ . OQ-C1 closed. ✓

## III-B. Unbounded Surprisal of the Bottom Element

### Lemma L-INF — Unbounded Surprisal of $\perp$

The surprisal  $I(n) = n$  at ball-hierarchy depth  $n$  is unbounded above: for any finite bound  $M$ , there exist depths  $n$  with  $I(n) > M$ .

$\perp = c_0$  corresponds to the limit point  $0 \in Q_2$  — the limit of the binary ball hierarchy at infinite depth. The binary branching measure assigns equal probability mass at each branch level (D4), and the surprisal diverges without bound as depth increases (T2). At this limit, no finite bound  $M$  contains the informational content of  $\perp$ .

Proof: Let  $M \in \mathbb{R}$ . By the Archimedean property,  $\exists n \in \mathbb{N}$  with  $n > M$ . Then  $I(n) = n > M$ . Since  $M$  was arbitrary, surprisal is unbounded above. ✓

Formal content: surprisal is not bounded above by any real  $M$ .

### Lemma L-INF — Unbounded Surprisal of $\perp$

Semantic content:  $\perp$  is informationally extreme — it is the limit point of the binary ball hierarchy under the 2-adic metric, the accumulation point approached by sequences of increasing depth. No finite bound  $M$  contains the surprisal at that limit; therefore no finite external interpreter can hold  $\perp$  as a static description. This is one argument for DA-1's precondition (ZP-E § I-DA1, Path 2), and it does not derive it. ZP-A CC-2 ( $\perp = \{\perp\}$ ), itself a commitment, gives a second, structural argument for the same conclusion: a self-containing object has no external interpreter by structure (ZP-A R3). The informational argument from the ball hierarchy and the structural argument from self-containment are independent; neither makes the configuration at  $P_0$  executing.

Note: L-INF supplies unbounded surprisal; the step from there to executing is not a consequence of L-INF and is not DA-1. It is a bridge principle of its own (ZP-E § I-DA1, Path 2), which the framework does not adopt, which is not the occurrence commitment, and which is not a premise of the Snap.

Status: DERIVED from D4 and T2. Structural corroboration: ZP-A CC-2 ( $\perp = \{\perp\}$ ) and R3. Lean: `l_inf` (purity check: no non-Mathlib axioms).

### Remark R-BRIDGE — On the Relationship Between $K$ and 2-Adic Surprisal

ZP-C uses two complexity measures: Kolmogorov complexity  $K(x|n)/|x|$  (Section I) and 2-adic surprisal  $I(n) = n$  (Section III). These measure related but distinct structures. Their relationship warrants explicit statement.

What each measures.  $K(x|n)/|x| = 1$  is a property of a specific binary string  $x$ : no program shorter than  $x$  generates it. It is a pointwise statement about a particular configuration at the incompressibility threshold.  $I(n) = n$  is a property of position in the ball hierarchy: the surprisal of any string at 2-adic depth  $n$  is  $n$ , by the branching measure RP-2. It is a structural property of depth, not of any particular string.

Where they coincide. At  $P_0$ , both conditions hold simultaneously for the same configuration:  $K(c_1|n)/|c_1| = 1$  (no shorter external description exists) and  $I(n) = n \rightarrow \infty$  (surprisal grows without bound as depth increases). This is not a coincidence of definition. The 2-adic limit point  $0 \in Q_2$  is the unique accumulation point of the binary ball hierarchy; algorithmic incompressibility is the  $K$ -complexity characterisation of the same extremality. Both locate the same threshold from independent directions. The structural argument for why this convergence is not accidental — that these two independently defined measures are forced to meet at the same limit by the framework's architecture — is developed in ZP-E Remark R-AFA.

Where they diverge. Away from  $P_0$ , the measures are not interchangeable.  $K(2^n) = O(\log n)$ : the  $n$ -fold power of 2 is compactly described by a short program, yet  $v_2(2^n) = n$  grows without bound. A string can be 2-adically deep without being Kolmogorov-incompressible, and vice versa. The two measures agree at the limit but diverge throughout the interior.

How ZP-C uses them. ZP-C uses  $K$  and  $I$  as independent convergent routes to the same conclusion — that  $\perp$  at  $P_0$  admits no finite external interpreter — not as a single unified measure. L-INF is the 2-adic surprisal argument:  $I(n)$  unbounded  $\rightarrow$  no finite interpreter can hold  $\perp$ . DA-1 Path 3 (ZP-E) is the  $K$ -incompressibility argument:  $K = 1$  rules out a shorter external generator, not an inert string, so Path 3 shows that executing is not derivable from incompressibility. Their independence is preserved deliberately. The convergence of two structurally distinct complexity measures on the same threshold is part of the argument for DA-1's precondition, which neither derives, and not a circularity within it. Axiomatising an equivalence between  $K$  and  $I$  would merge these two routes into one and eliminate that independence.

## IV. Execution as State: The Hardware Lemma

### Definition D7 — Machine Configuration

A machine configuration  $c$  is a complete description of a Turing machine at a given moment: the current state of the control unit, the position of the read/write head, and the contents of all tape cells.

$c_0$ : the initial configuration — the machine exists but has not yet begun execution (no instruction has been fetched).

$c_1$ : the first running configuration — the machine has fetched and begun executing its first instruction.

Note: D7 does not import physics. It is a structural distinction between two discrete machine states within the standard Turing model.

### Lemma L-RUN — Execution is a Non-Null State Change

The transition from  $c_0$  (initial: not yet running) to  $c_1$  (running: first instruction fetched) is a discrete, irreducible state change in the machine configuration.

Proof:

Step 1 —  $c_0$  and  $c_1$  are distinct configurations by D7: the control unit is in a different state (idle vs. executing).  $c_0 \neq c_1$ .

Step 2 — By AX-B1, a state either exists or it does not. The machine at  $c_1$  occupies a configuration distinct from  $c_0$ . The transition  $c_0 \rightarrow c_1$  is a binary state change in the sense of AX-B1.

Step 3 — The transition is irreducible: there is no intermediate configuration between  $c_0$  and  $c_1$ . The first instruction fetch is the minimal unit of execution.

Step 4 —  $c_0$  is modeled as corresponding to  $\perp$  in the semilattice (CC-2 — see below).  $c_1$  is strictly above  $\perp$ : the machine configuration has gained content (an active execution context) not present in  $c_0$ .

Conclusion:  $c_1 \neq \perp$ . The act of execution is itself a nonzero state, regardless of what the output tape contains. ✓

Status: DERIVED from AX-B1 and D7. No Coding Theorem required. No output-tape contents required.

### Conditional Claim CC-2 — $c_0 = \perp$ (Parallel to CC-1 in ZP-A)

We model the initial machine configuration  $c_0$  as corresponding to  $\perp$  in the semilattice  $(L, \vee, \perp)$ . This is not derived from D7 or from A1–A4 — it is a modeling choice.

Motivation:  $c_0$  is the machine state prior to any execution: no instruction has been fetched, no content has accumulated.  $\perp$  is the algebraic element below all others — the state of no accumulated content. The identification is motivated by structural correspondence, not derivation.

Parallel: CC-1 in ZP-A commits to  $S_0 = \perp$  for state sequences. CC-2 makes the same commitment for machine configurations. Both are modeling choices, not derivations from the core axioms.

Effect: Step 4 of L-RUN depends on CC-2. The conclusion  $c_1 \neq \perp$  follows from L-RUN's proof; the identification of  $c_0$  with  $\perp$  is the additional modeling commitment supplied by CC-2.

Status: CONDITIONAL CLAIM — modeling commitment; not derived from D7 or A1–A4.

| Remark R4 — Configuration and Output Are Independent                                                                                                                                                           |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| D-TQ-2 (Output Definition): A computation's output is defined by the contents of its tape/register at termination.                                                                                             |
| L-RUN establishes that configuration state and output state are independent. A program may terminate with null output while still having passed through non-null configuration states during execution.        |
| The question "can a program output $\perp$ without any non-null intermediate state?" is answered by looking at the execution trace, not the output tape. This distinction is load-bearing for T-BUF and TQ-IH. |

## V. The Test Question and the Buffer Overflow Theorem

| Test Question TQ-IH — Can a Program Output $\perp$ Without a Non-Null Intermediate State?                                                                                                    |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Question: Does there exist a program $p$ such that $U(p) = \perp$ and the execution trace $\tau(p)$ contains no configuration $c_i$ with $c_i \neq \perp$ ?                                  |
| Answer: No.                                                                                                                                                                                  |
| Proof:                                                                                                                                                                                       |
| Step 1 — Any program that executes must pass through $c_1$ (by definition of execution, D7).                                                                                                 |
| Step 2 — By L-RUN, $c_1 \neq \perp$ .                                                                                                                                                        |
| Step 3 — Therefore the execution trace $\tau(p) = (c_0, c_1, \dots)$ contains $c_1$ , which is a non-null configuration state, regardless of what appears on the output tape at termination. |
| Step 4 — The output of $p$ being $\perp$ does not imply that all configurations in $\tau(p)$ are $\perp$ . Output state and configuration state are distinct (R4).                           |
| Conclusion: No program can produce $\perp$ as output without passing through a non-null intermediate configuration state. TQ-IH answered negatively. ✓                                       |
| Status: DERIVED from L-RUN, D7, and R4. No Kolmogorov machinery required. TQ-IH is one of the results from which ZP-E derives the shape of the Snap as Theorem T-SNAP.                       |

| Remark R-TQ — External Confirmation of TQ-IH (April 2026)                                                                                                                                                                                                                                                                                                                               |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| TQ-IH was independently reviewed by an external PhD mathematician.                                                                                                                                                                                                                                                                                                                      |
| Confirmation: "As long as you have $c_1 \neq c_0$ , this implies $c_i \neq c_0$ for some $1 \leq i < n$ , at least for $n \geq 2$ . This has nothing to do with computation or anything at all, it's a triviality."                                                                                                                                                                     |
| This confirms two things. First, TQ-IH holds. Second, and more significantly, the result is domain-independent: it does not depend on any Turing-specific machinery, any Kolmogorov complexity argument, or any property of the semilattice. It holds in any setting where a non-null first step $c_1 \neq c_0$ is given. The derivation from L-RUN alone is complete and unassailable. |
| Status: EXTERNALLY CONFIRMED — April 2026.                                                                                                                                                                                                                                                                                                                                              |

### Candidate Theorem T-BUF — Incompressibility Forces Non-Null Execution State

Statement: At the incompressibility threshold  $P_0$ , the Binary Snap  $\perp \rightarrow \epsilon_0$  is a structural consequence of execution, not an external trigger.

Step 1 —  $P_0$  identifies the configuration  $x$  at which  $K(x|n)/n = 1$ : the configuration string is incompressible. (D1)

Step 2 — The configuration at  $P_0$  is executing: it is a running machine's current configuration (Sense B, ZP-E § I-DA1), not an inert string. This step is not derived: it is the occurrence commitment (instantiation occurs), and T-BUF is conditional on it. Arguments for it, none deriving it: the configuration is informationally extreme (L-INF), and the step from unbounded surprisal to executing is a bridge principle of its own, which the framework does not adopt (ZP-E, DA-1 Path 2); it is incompressible (D1), which rules out a shorter external generator and not an inert string (ZP-E, DA-1 Path 3). DA-1 (ZP-E § I-DA1, closed given DP-2) consumes this step and does not supply it. In ZP-A's order, given CC-1, its first-step form is the hypothesis  $\text{hocc}$  of  $t\_snap\_given$  (ZP-E, Premises of T-SNAP).

Step 3 — Any execution passes through  $c_1$  (L-RUN).  $c_1 \neq \perp$  (L-RUN conclusion).

Step 4 — In  $(L, \vee, \perp)$ , this non-null configuration state is  $c_1 = \perp \vee \epsilon_0$ . By ZP-A D2, this is the Binary Snap.

Conclusion: Given Step 2, execution at  $P_0$  guarantees a non-null configuration state (L-RUN), and that state is  $\epsilon_0$  in the semilattice. The derivation pathway is  $P_0 + \text{L-RUN} + \text{TQ-IH} + \text{ZP-A D2}$ , given Step 2. ✓

Status: CANDIDATE THEOREM — conditional on execution. Steps 1, 3 and 4 are derived within ZP-C; Step 2 is the occurrence commitment (ZP-E § I-DA1), not a derived step, and DA-1 does not close it. ZP-E derives the Snap's shape as Theorem T-SNAP; that the Snap occurs follows from the occurrence commitment together with DA-1 (closed given DP-2).

### Remark R5 — Updated Status of AX-1

AX-1 (Binary Snap Causality) was previously labeled Axiomatic in ZP-C.

AX-1 was a Candidate Theorem in ZP-C; it is now retired: its shape is proved as T-SNAP (ZP-E), and that the Snap occurs is stated separately: it follows from the occurrence commitment (instantiation occurs) together with DA-1 (closed given DP-2). The derivation pathway ZP-C gave it:  $P_0$  (D1) identifies the threshold. L-RUN establishes that execution at the threshold constitutes a nonzero state change. TQ-IH establishes that no program avoids this. ZP-A D2 establishes that a nonzero state change from  $\perp$  is the Binary Snap.

That a configuration at  $P_0$  is an execution event (a running machine's current configuration, not an inert string) is the occurrence commitment, DA-1's precondition. DA-1 (ZP-E, a Derived Proposition closed given DP-2) consumes it and does not supply it: given it,  $c_0 \rightarrow c_1$  is D7's. Both bear on whether the Snap occurs, not on its shape.

Status label: RETIRED in the ZP-E DA-1 insert, as stated above. AX-1 was a Candidate Theorem in ZP-C, with its gap identified: T-BUF Step 2, which is the occurrence commitment.

## VI. Open Items Register

| Item                         | Status      | Description                             |
|------------------------------|-------------|-----------------------------------------|
| S1: Distribution stipulation | Closed — T1 | T1 derives P and Q from AX-B1 and RP-1. |

| Item                                             | Status                            | Description                                                                                                                                                                                                          |
|--------------------------------------------------|-----------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| OQ-C1: Non-conservation                          | Closed — T2 rebuilt               | Telescoping critique resolved; infinite divergence proven within extended D6.                                                                                                                                        |
| Smooth embedding, MO-1, P1                       | Retired                           | Retired. Inconsistent with ZP-B.                                                                                                                                                                                     |
| RP-1: Representation principle                   | Principle — explicit              | Bridge between AX-B1 and probabilistic tools.                                                                                                                                                                        |
| RP-2: Branching measure on $Q_2 \setminus \{0\}$ | Principle — explicit              | Canonical branching measure; representational commitment; required by T2 and L-INF.                                                                                                                                  |
| CC-2: $c_0 = \perp$                              | Conditional Claim                 | Modeling commitment — $c_0$ identified with $\perp$ in semilattice. Parallel to CC-1 in ZP-A. Required by L-RUN Step 4.                                                                                              |
| D7: Machine configuration                        | Defined                           | Foundation for L-RUN and TQ-IH.                                                                                                                                                                                      |
| L-RUN: Hardware Lemma                            | Derived — Lemma                   | Execution is a nonzero state change. Derived from AX-B1 and D7.                                                                                                                                                      |
| TQ-IH: Test Question                             | Closed — Confirmed                | No program can output $\perp$ without a non-null intermediate configuration state. Proven by L-RUN. Externally confirmed April 2026 (R-TQ): domain-independent, requires no Turing-specific or Kolmogorov machinery. |
| T-BUF: Buffer Overflow Theorem                   | Candidate Theorem                 | Conditional on execution: given that the configuration at $P_0$ is executing (Step 2, the occurrence commitment), its state is non-null and is $\epsilon_0$ . DA-1 consumes Step 2 and does not supply it.           |
| AX-1: Binary Snap Causality                      | Retired (was a Candidate Theorem) | Derivation pathway formalized. AX-1 retired in the ZP-E DA-1 insert: shape proved as T-SNAP; the snap occurs given the occurrence commitment and DA-1.                                                               |

## VII. Validation Status

| Component                             | Status / Notes                                                                                                                                                        |
|---------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $K(x n)$ and $P_0$ (D1)               | Valid — standard algorithmic IT                                                                                                                                       |
| R1: Scope of $P_0$                    | Valid — updated: AX-1 was a Candidate Theorem, not a bare axiom; it is now retired (shape proved as T-SNAP; the snap occurs given the occurrence commitment and DA-1) |
| RP-1: Representation Principle        | Valid — Principle; explicit bridge; resolves reviewer gap in T1                                                                                                       |
| T1: Distributions from AX-B1 + RP-1   | Valid — Derived; reviewer gap closed                                                                                                                                  |
| D2: JSD definition                    | Valid — standard                                                                                                                                                      |
| T1b: $E = \text{JSD} = 1 \text{ bit}$ | Valid — Derived from AX-B1 + RP-1                                                                                                                                     |

| Component                                 | Status / Notes                                                                                                                                                                                         |
|-------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| D3: Dirac measure $\delta_0$              | Valid — standard; compatible with discrete $Q_2$ topology                                                                                                                                              |
| R3: Smooth embedding retired              | Valid — Retired; inconsistent with ZP-B                                                                                                                                                                |
| D4: Discrete surprisal $I(x)$             | Valid — pointwise on $Q_2 \setminus \{0\}$ ; branching measure stated                                                                                                                                  |
| RP-2: Branching measure                   | Valid — Principle; explicit representational commitment; canonical binary branching measure                                                                                                            |
| CC-2: $c_0 = \perp$                       | Valid — Conditional Claim; modeling commitment; parallel to CC-1 in ZP-A; load-bearing for L-RUN Step 4                                                                                                |
| D5: Difference operator DF                | Valid — antisymmetric; no smoothness assumed                                                                                                                                                           |
| D6: Circulation (extended)                | Valid — finite and infinite cases both defined; finite conservation acknowledged                                                                                                                       |
| T2: Non-conservation rebuilt              | Valid — Derived; telescoping critique addressed; divergence at $\sigma \rightarrow 0$ established                                                                                                      |
| R2: Hamming cross-validation              | Valid — Consistency check                                                                                                                                                                              |
| D7: Machine configuration                 | Valid — Defined; standard Turing model; no physics imported                                                                                                                                            |
| L-RUN: Hardware Lemma                     | Valid — Derived from AX-B1 and D7                                                                                                                                                                      |
| R4: Configuration vs. output independence | Valid — load-bearing distinction for TQ-IH and T-BUF                                                                                                                                                   |
| TQ-IH: Test question answered             | Valid — Derived by L-RUN; no Kolmogorov machinery required                                                                                                                                             |
| T-BUF: Candidate Theorem                  | Candidate — conditional on execution; Steps 1, 3, 4 derived in ZP-C; Step 2 is the occurrence commitment (ZP-E), not closed by DA-1                                                                    |
| R5: AX-1 status updated                   | Valid — AX-1 was a Candidate Theorem in ZP-C (prior Axiomatic status corrected); it is now retired: shape proved as T-SNAP (ZP-E); the snap occurs given the occurrence commitment and DA-1            |
| R-BRIDGE: K vs. 2-adic surprisal          | Valid — Remark; states relationship explicitly: distinct measures converging at $P_0$ ; independence of L-INF and K paths preserved; cross-reference to ZP-E R-AFA for structural convergence argument |